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DEPARTMENT OF MATHEMATICS "TULLIO LEVI-CIVITA"

BACHELOR'S DEGREE IN COMPUTER SCIENCE



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Explainable Machine Learning
through Large Language Models:
Analysis and Prompt Design

Bachelor's Thesis

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Abstract

As Machine Learning (ML) models become increasingly prevalent in business applications, ensuring their transparency and reliability through Explainable Artificial Intelligence (XAI) is critical. This project, conducted in collaboration with Zucchetti S.p.A., investigates the interpretability and comprehensibility of various machine learning algorithms.

The research systematically evaluates a spectrum of predictive models, ranging from fundamental Regression (regression) and Classification (classification) models—such as Linear Regression, Logistic Regression, Support Vector Machines, and Decision Trees—to complex ensemble methods, including Random Forest, XGBoost, RuleFit, and Symbolic Regression. The analysis explores both intrinsic model transparency and post-hoc explainability techniques, utilizing metrics like SHapley Additive exPlanations (SHAP) to interpret predictions and feature importance. Furthermore, the methodology encompasses the examination of mathematical foundations, internal knowledge representations, and the impact of ensemble techniques on both model performance and explainability. Real-world validation is performed using datasets such as Student Salary Prediction.

A primary objective of this work is bridging the gap between technical accuracy and human-understandable explanations for non-expert stakeholders. To achieve this, the project leverages advanced Prompt Engineering principles. Tailored prompts for Large Language Model (LLM) are developed for each analyzed algorithm. These prompts are explicitly designed to automatically generate human-readable explanations of algorithmic behaviors and predictions. Ultimately, the project delivers production-ready implementations and LLM integration adapters, demonstrating how the synthesis of traditional XAI methodologies with modern language models can significantly enhance the transparency, trust, and responsible deployment of AI systems.

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Lorenzo Soligo

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Chapter 1.

Introduction

Introduction to the stage idea, goals, company and thesis organization. This chapter is meant to provide the necessary context for the reader to understand the motivation and the structure of the thesis.s

1.1. Company

Zucchetti S.p.A. is a leading Italian software company specializing in providing comprehensive IT solutions for businesses. Founded in 1978, Zucchetti has grown to become one of the largest software providers in Italy, offering a wide range of products and services that cater to various industries, including manufacturing, retail, healthcare, and public administration.

With a strong focus on innovation and customer satisfaction, Zucchetti is committed to helping organizations optimize their operations and achieve their business goals through cutting-edge technology. Exactly from this commit, the company has been actively involved in the development and implementation of ML and Artificial Intelligence (AI) solutions, aiming to enhance the capabilities of their software offerings and provide more value to their clients.



Figure 1: Zucchetti S.p.A. logo

1.2. Stage idea

The internship project is centered around the exploration of XAI techniques, with a specific focus on the interpretability and comprehensibility of machine learning algorithms. The primary objective is to analyze a variety of predictive models, ranging from basic regression and classification algorithms to more complex ensemble methods, in order to evaluate their transparency and explain-

ability. The idea is to increase the understanding of the results of the models, and to make them more accessible to non-expert stakeholders, by leveraging advanced Prompt Engineering (PE) principles to develop tailored prompts for Large Language Models that can automatically generate human-readable explanations of algorithmic behaviors and predictions.

The timeline of the project is structured as follows:

Week	Task
1st	Analysis of Linear Regression, Logistic Regression, Support Vector Machines.
2nd	Analysis of Decision Trees, Random Forests, XGBoost.
3rd	Code implementation and Prompt Engineering for the analyzed algorithms.
4th	Evaluation and documentation of the results.
5th	Analysis of RuleFit and deep dive in Random Forests.
6th	Analysis of Symbolic Regression.
7th	Code implementation and Prompt Engineering for the analyzed algorithms.
8th	Evaluation and documentation of the results, final report writing.

Table 1: Table of the project timeline

1.3. Project goals

The goals follow a notation to distinguish between necessary (N), desirable (D) and optional (O) goals. In the planning phase the goals were defined as follows:

Goal	Description
O-01	Complete and profound comprehension of the choosen algorithms, their mathematical foundations and their internal knowledge representation.
O-02	Comprension of interpretability and explainability techniques in machine learning.
O-03	Comprehension of algorithms design techniques.
O-04	Prompt generation for Large Language Models to generate human-readable explanations.
O-05	Comprehension and application of Prompt Engineering principles in the context of XAI.
D-01	Creation of a metric to evaluate the explainability of the analyzed algorithms.
D-02	Automatization of the analysis pipeline from dataset to LLM requests.
F-01	Application in real-world scenarios of the interpretability and explainability of Machine Learning models.
F-02	Explainability study of semi-supervised and unsupervised algorithms.

Table 2: Table of project goals

1.4. Thesis structure

The second chapter describes the basic concepts that fuel the project. The concepts spread from the mathematical foundations of the analyzed algorithms, the basics of explainability, to the principles of Prompt Engineering and LLMs.

The third chapter analyzes the ML algorithms, focusing on their capability, limitation and interpretability.

The fourth chapter describes the design and implementation of the analysis pipeline from dataset to LLM requests.

The fifth chapter discusses prompt engineering techniques and their application in the context of XAI.

The sixth chapter presents the conclusions of the study, the findings and the goals assessment.

Chapter 2.

Background knowledge

In this chapter, we will provide the necessary background knowledge to understand the project. We will cover the mathematical foundations of the analyzed algorithms, the basics of explainability, and the principles of Prompt Engineering and LLMs.

2.1. Algorithmic analysis structure

The analysis of the algorithms is structured in a systematic way to ensure a comprehensive evaluation of their interpretability and comprehensibility as well as their functioning. The structure includes the following components:

1. **Mathematical foundations:** detailed examination of the mathematical principles underlying each algorithm, including their assumptions, optimization techniques, and theoretical properties. The focus is on the understanding of the specific Objective Function (objective function) that the algorithms optimize, the regularization techniques they use, and the mathematical properties that govern their behavior.
2. **Computational Complexity (computational complexity):** analysis of the computational requirements of each algorithm, including time complexities and scalability regarding both training and inference.
3. **Internal representation:** exploration of how each algorithm internally represents data and makes decisions, especially regarding Categorical Features (categorical features).
4. **Data assumptions:** investigation into the assumptions each algorithm makes about the data, such as linearity, independence of features, and distributional assumptions.
5. **Predictive performance and limitations:** evaluation of the predictive performance of each algorithm on relevant datasets, considering the data assumptions and mathematical foundation of the model.
6. **Metrics for prediction quality:** presentation of the most relevant metrics for evaluating the models performance, based on the task.
7. **Metrics for interpretability:** presentation of the most relevant metrics for evaluating the models interpretability, based on the task.
8. **Explainability limitations:** discussion of the limitations of explainability techniques for each algorithm, including potential pitfalls and challenges in interpreting their predictions.

2.2. Explainability

Explainability (or interpretability) of a ML model is the ability to communicate **how and why** the model made a prediction.

Contrary to intuition, explainability it's **not a binary attribute**, but a spectrum with multiple dimensions.

2.2.1. Explainability dimensions

1. Intrinsic Transparency (Model-Level Interpretability)

The model is inherently transparent and interpretable by design. The structure directly communicates how it works, such as regression coefficients or decision tree nodes. Each prediction is fully traceable, and parameters have tangible meanings. Explanations reflect human reasoning, making them coherent from a domain perspective.

It represents the best case scenario for explainability but it is often in trade-off with performance, as more complex models tend to be less transparent but more powerful. It is generally organized in tiers: High, Medium and Low transparency.

2. Global Interpretability (Model-Level Explainability)

The model is comprehensible in its entirety, allowing to understand its overall behaviour and decision-making process. Such a level of understanding it's hardly achievable and it is usually indirectly reached by comprehending the singular components of the models instead of the model as a whole.

3. Modular Interpretability

When global interpretability is not achievable, we can still understand the model by analyzing its components separately. Exploiting the model transparency, we can understand the role of each component in the decision-making process.

This concept is clear in the case of linear methods, where it is easy to understand the contribution of the single weights or in tree based models, where the splits and the structure of the tree can be analyzed separately.

4. Local Interpretability (Instance-Level Explainability)

The model might be opaque and too complex to understand globally. However, for a specific prediction, we can generate an explanation that is locally interpretable. SHAP could be used in this context to explain a particular instance obtained a certain prediction.

2.2.2. Explanation properties and factors that facilitate understanding

For human explanations to be effective, they should have certain properties that facilitate understanding and usability. The following properties can be identified from the work of Robnik-Sikonja and Bohanec 2018 and Molnar [1]:

- **Contrastive explanations**

Comparative explanations that highlight the differences between instances can be more informative than absolute explanations. The instances confrontation makes the explanation more impactful, providing a direct idea of how different factors contribute to different outcomes.

- **Cause selection**

Not all features are equally important; explanations should focus on the most influential factors rather than listing all contributing features. A small sample of the most relevant features provide a more clear explanation than a long list of all features preventing information overload.

- **Social and contextual factors**

The social environment and the audience’s expertise level should be considered when generating explanations to ensure they are appropriate and understandable.

- **Anomalies or abnormal predictions**

Highlighting anomalies or abnormal predictions can help users understand when the model is making an unusual decision, which can be crucial for trust and debugging. Features that are not largely relevant for the majority of the predictions but that have a large impact on specific predictions should be highlighted for their capability.

- **Fidelity**

The explanation should accurately reflect the model’s behavior and not introduce misleading information. A high-fidelity explanation ensures that users can trust the insights provided and make informed decisions based on them, provided that the model is accurate in its predictions.

- **General and probable**

In the absence of anomalies of specific cases, a good explanation should rely on general and probable causes that are likely to be true for most instances. This parameter can be quantified by the feature support:

$$\text{Feature Support}_i = \frac{\text{\#training instances with similar values to } x_i}{\text{\#total training instances}}$$

Generality can be misleading, as it may lead to explanations that are too broad and not specific enough to be useful. It’s important to balance generality with the impact of the explanation, ensuring that it provides meaningful insights without being overly vague. This is why usually abnormal features are highlighted, as they can provide more specific and impactful explanations.

2.2.3. SHAP analysis

SHAP (SHapley Additive exPlanations) is a method for explaining the output of machine learning models by attributing the prediction to each feature. It is based on the concept of Shapley values from cooperative game theory, which provide a way to fairly distribute the “payout” (in this case, the prediction) among the “players” (the features) based on their contribution to the prediction. SHAP values can be used to understand the importance of each feature for a specific prediction (local interpretability) or for the model as a whole (global interpretability). SHAP provides a unified framework for interpreting various types of models and can be applied to both linear and non-linear models, making it a powerful tool for explainable AI.



Figure 2: SHAP library logo

2.3. Prompt Engineering and LLMs

The following sections will present the principles used in the design of the prompts for the Large Language Models to generate human-readable explanations of the analyzed algorithms.

2.3.1. Expert Personas

The expert personas techniques consists in the design of prompts that describes the role that the LLM should play in the task. This technique first should guide the model in generating a more precise, coherent and relevant output.

Though the expert personas seems to be linked to a loss of accuracy in multiple benchmarks in the recent studies by Zizhao Hu [2], the principal use of the LLMs in this project is results evaluation and text generation. The data and metrics do not have to be generated and have only to be read directly from the prompt. The paper specifically talks about “*For tasks that depend on pretrained knowledge retrieval accuracy*” [2]. It’s reasonable to think that in such a context, loss of accuracy is not a critical issue. In the same paper, an alignment of behavior and style to the described personas is observed. This drift in behavior is critical in the task of describing the results of an algorithm in a human-readable way, as the style of the explanation is fundamental to make it accessible to non-expert stakeholders.

2.3.2. Familiar formats

The extensive use of Large Language Models have led to the emergence of some familiar formats that are widely used in the design of prompts.

The principal features that this formats present, and that makes them particularly effective, are the following:

- Widely use in training: the more a format is used in the training of the model, the more likely is to be effective in the interpretation of content in that format.
- Clear structure: the format should have a clear and recognizable structure that allows the model to easily identify the different components of the prompt and their relationships.
- Token efficiency: the format should be designed in a way that minimizes the number of tokens used, while still conveying all the necessary information.

This is particularly important in the context of LLMs, as the number of tokens used can have a significant impact on the performance and cost of the model.

From the multitude of formats in existence, some emerged for their particular accuracy and efficiency. The most valueable one is **Markdown**, which has a clear and recognizable structure that allows the model to easily identify the different components of the prompt and their relationships. The use of mark-down also allows to create a clear and organized structure for the generated explanations, making them more readable and accessible to non-expert stakeholders.

Another accurate format is YAML, which offers a more readable formatting than JSON and XML while still being well structured.

An alternative like CSV can be used, as it is a widely used format for tabular data and is likely to be well understood by the model. However, it's important to ensure that the CSV format is properly structured and that the necessary information is included in the prompt to allow the model to generate accurate and relevant explanations.

What follows is a summarizing image where the 11 formats taken in exam in the study[3] are presented. The y-axis is a scale of Accuracy while the x-axis a token consumption scale. It's clear that in this particular case of study, only OpenAI's GPT-4.1 nano has been tested, Markdown emerges as the most accurate format, while maintaining good token usage.

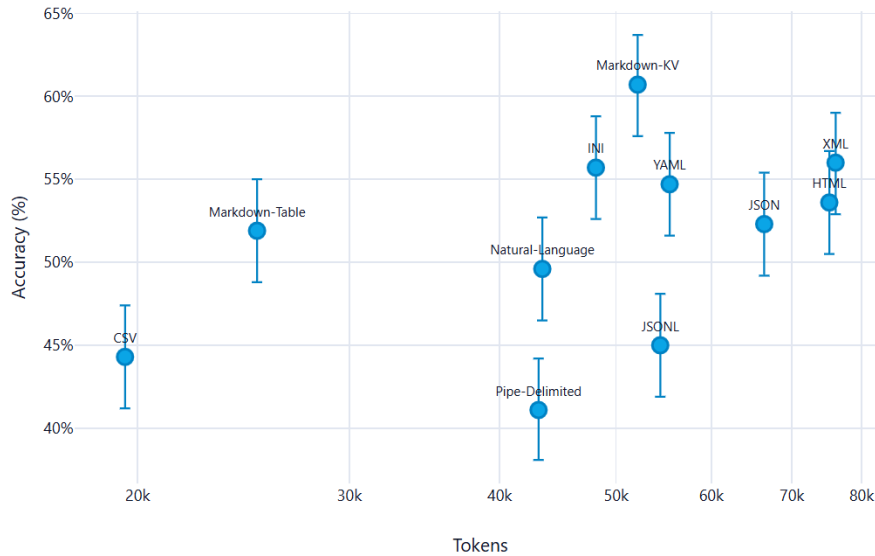


Figure 3: Formats for prompting. Source: @formats-llms

2.3.3. Chain-of-Thought

Chain of Thought (CoT) prompting is a technique that consists in the design of prompts that guide the Large Language Models to generate a step-by-step reasoning process to reach the final answer. This technique is particularly useful in the context of this project, as it allows to generate explanations that are

more coherent and relevant to the analyzed algorithms.

The additional value of the guided, step-by-step reasoning has been proven in multiple benchmarks[4], and it is particularly useful in the context of this project, as it allows to generate explanations that are more coherent and relevant to the analyzed algorithms.

In the particular context of XAI and non-technical audiences because it enhances the model's ability to provide detailed and understandable explanations. The generated answer is not just a final result, but a comprehensive explanation that breaks down the reasoning process into clear and logical steps.

2.3.4. Zero-Shot prompting

This technique consists in the design of prompts that do not include any example of the expected output. It's the simplest kind of prompting because it erases the problem of having to provide training examples. Such method is particularly necessary in the context due to the fact that an example of data and metrics for reference could be too specific and not generalized enough. An additional set of data could also confuse the model and make it less accurate or biased on the actual result of the analysis.

The zero-shot prompting also revealed to be quite effective with some additional techniques such as the use of Chain of Thought (CoT) prompting [5].

2.3.5. Multimodal prompt

The multimodal prompting technique consists in the design of prompts that include multiple modalities, such as text, images, tables, etc. This technique is particularly useful in the context of this project, as it allows to provide the model with a richer and more comprehensive context for generating explanations.

The use of multimodal prompts can enhance the model's ability to comprehend and analyze the results of the algorithms, as it can leverage the information provided in different formats to generate more accurate and relevant explanations. In fact, *“visual prompts effectively interact with the textual prompts, enhancing the alignment between modalities and thereby improving the model's performance on the zero-shot instruction learning”* [6].

Chapter 3.

ML algorithms analysis

Breve introduzione al capitolo

3.1. Linear regression

3.2. ...

Chapter 4.

Design and code implementation

Breve introduzione al capitolo

4.1. Tecnologie e strumenti

Di seguito viene data una panoramica delle tecnologie e strumenti utilizzati.

4.1.1. Tecnologia 1

Descrizione Tecnologia 1.

4.1.2. Tecnologia 2

Descrizione Tecnologia 2

4.2. Progettazione

4.3. Design Pattern utilizzati

4.4. Codifica

Chapter 5.

Prompt Engineering

Breve introduzione al capitolo

Chapter 6.

Conclusion

6.1. Consuntivo finale

6.2. Raggiungimento degli obiettivi

6.3. Conoscenze acquisite

6.4. Valutazione personale

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Chapter 7.

Glossary

AI – Artificial Intelligence: Field of computer science that focuses on creating systems capable of performing tasks that typically require human intelligence, such as learning, reasoning, problem-solving, and understanding natural language. 1.

categorical features – Categorical Features: Features in a dataset that represent discrete categories or groups, rather than continuous numerical values, and often require special handling in machine learning algorithms. 4.

classification – Classification: A type of machine learning problem in which the goal is to predict discrete class labels based on input features. i, 1.

computational complexity – Computational Complexity: A measure of the amount of computational resources (time and space) that an algorithm requires to run, often expressed in terms of the size of the input data. 4.

XAI – Explainable Artificial Intelligence: A field of artificial intelligence that focuses on creating models and systems that can provide clear and understandable explanations for their decisions and actions. i, i, 1., 9.

LLM – Large Language Model: A type of artificial intelligence model that is trained on vast amounts of text data to understand and generate human-like language. i, i, 2., 7., 7., 7., 8.

ML – Machine Learning: A subset of artificial intelligence that involves training algorithms to learn patterns from data and make predictions or decisions without being explicitly programmed. i, 1.

objective function – Objective Function: A mathematical function that an algorithm optimizes during training, which defines the goal of the learning process and guides the model's adjustments to minimize error or maximize performance. 4.

PE – Prompt Engineering: The process of designing and optimizing prompts to effectively communicate with language models and elicit desired responses. 2.

regression – Regression: A type of machine learning problem in which the goal is to predict continuous numerical values based on input features. i, 1.

SHAP – SHapley Additive exPlanations: A method for explaining the output of machine learning models by attributing the prediction to each feature. i